

Project 2 Debrief — transcript

Started: 2026-09-26T23:23:07-05:00
Completed: 2026-09-26T23:48:05-05:00
Elapsed wall time: 25 minutes
Timing note: Elapsed time includes pauses between messages.
Tool: Claude Code desktop
Model: Opus 5.5
Effort setting: Medium
User messages: 43
Agent messages: 42

- **Project:** ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator
- **Repository:** <https://github.com/rg-sureshkumar-goat/AI-Footprint-Calculator>
- **Branch:** project-2 (HEAD at time of debrief: 8d56084)
- **Learner:** RG Sureshkumar
- **Protocol:** debrief.md (repo root)

Note on redaction. Sanitised following transcripts/README.md: the macOS home directory path and username, the author's email address, and temporary session paths are replaced. No messages were removed or reworded.

Note on tool calls. The agent ran read-only inspection throughout (reading debrief.md, brief.md, research.md, spec.md, plan.md, the stage transcripts, and core.js; git log; date; and one web fetch of source S1). These are summarised in square brackets rather than reproduced in full; the agent's internal reasoning is not included. No project files were modified and no build or development commands were run. The only file written was this transcript (and its PDF export).

Note on message counts. Counted from the learner's opening debrief prompt through the learner's direction to save the transcript. Tool calls, tool output, and the agent's confirmation after saving are not counted.

1. Learner (RG Sureshkumar)

Read ./debrief.md in full and follow it as the authoritative protocol. Begin the Project 2 debrief now. If you cannot locate or read that file, say so and stop.

Agent

[Tool: Bash — read debrief.md in full (217 lines)] [Tool: Bash — system clock: 2026-09-26T23:23:07-05:00]

I've read debrief.md in full. The setup checks out: the file is in the repository root, and the project-2 branch is checked out.

Welcome to Project 2 Debrief. Let's talk through how you researched, built, and checked your five features for Project 2. If something's unfamiliar, just say so and I'll explain it. This will take roughly 30–60 minutes.

Session started: 2026-09-26T23:23:07-05:00. Elapsed time will include pauses between messages.

To start, which tool, model, and effort setting are you using for this conversation? "Not sure" is a fine answer.

2. Learner (RG Sureshkumar)

Claude Code desktop, Opus 5.5, Medium

Agent

[Tool: Bash — list repository files, document sizes, and git log main..project-2 (23 commits: stage documents, six checkpoint commits 6b57f78, 397e7eb, 3210da9, c6624c1, e793423, 87fdd88, and transcript updates)] [Tool: Read — brief.md, spec.md, plan.md in full] [Tool: Bash — research.md headings, reference profiles, and audience needs §1–2]

Thanks: Claude Code desktop, Opus 5.5, Medium. I've now read the brief, spec, plan, and the start of the research, and looked at the branch history.

Let's start where the project started. You branched project-2 from your submitted Project 1 commit, and you'd already done a Project 1 debrief. Did anything from that debrief change how you approached Project 2? Give me one concrete example if you can.

3. Learner (RG Sureshkumar)

nothing from the previous debrief changed my approach to project 2

Agent

That's a fine answer. Let's look at how you recorded progress instead. Your branch has one commit per feature (for example 397e7eb for images and 3210da9 for agent sessions), and each one was made after you verified that checkpoint. What did those checkpoint commits give you? For instance, if feature 4 had broken something, what could you have done?

4. Learner (RG Sureshkumar)

I could have gone into the actual project and test out the feature manually and tell claude what the problem was and suggest ways to fix it

Agent

Testing it yourself and telling Claude what went wrong is a good habit. But that answer would hold even without commits. What commits add is a known-good point you can return to. If feature 4 had broken feature 3, you could compare the code with c6624c1, or go back to it, instead of untangling it all by hand.

Let's move on to research. How did you get from the brief and the three profiles to your five features? Which alternative did you seriously consider and then drop?

5. Learner (RG Sureshkumar)

I has claude recommend me some features based on my requirements and then I narrowed them down based on what I knew I needed

Agent

[Tool: Bash — read research.md "Possible features" (12 ideas) and "Selected features"]

That's a reasonable start: Claude generated twelve options and you chose from them. The interesting part is "what I knew I needed." Your research notes say "ranges carried through every result" was the best-evidenced answer to Robin's trust concern, but you chose like-for-like comparisons as your fifth feature. Why that one over ranges?

6. Learner (RG Sureshkumar)

Because I wanted a feature that showed me comparisons and the ranges looked like another feature that I already had

Agent

That fits your spec, which says ranges were already built into features 1–4. But notice the wording: "I wanted a feature that showed me comparisons." The brief asks for the *strongest remaining need*, meaning an employee's need rather than yours. Which profile needed like-for-like comparisons most, and what in the research showed that need?

7. Learner (RG Sureshkumar)

Im not sure

Agent

That's worth untangling. A **user need** is what a person needs to understand or do. A feature is only one possible answer to it. Your research actually made the case: Alex cares about water, and the old burger comparison used mostly rainwater, so it didn't match AI water. Robin, who is skeptical, would catch a comparison like that. Does that match why the feature ended up mattering?

8. Learner (RG Sureshkumar)

yes it does

Agent

[Tool: Bash — read research.md source assessments S1–S3 and the S59–S80 assessment notes]

Good. Let's follow one number to its source. Feature 1 counts each image as 1.35 Wh, taken from S1, Luccioni et al. (2024). What did you check yourself in that paper before accepting 1.35? For example, did you open it, and what exactly did it measure?

9. Learner (RG Sureshkumar)

It measured the per unit power that it required for image generation

Agent

Close. Two details matter, though. First, it measured **energy** (watt-hours), not power (watts). Second, the paper reports 1.35 kWh per *1,000 images*, from 2022–23 open-source models on one research GPU, not from tools like Midjourney. Did you open the paper yourself, or did you rely on Claude's summary of it?

10. Learner (RG Sureshkumar)

i used claudes summary

Agent

Thanks for being straight about that. A summary tells you what Claude says the source contains; it doesn't show you what the source actually contains. That gap is where **hallucination** gets in: a misread unit, a wrong table, or a paper that doesn't exist. Here the move would have been to open the arXiv link, find Table 2, and confirm the 1.35 and its units yourself.

How many of your sources did you open directly?

11. Learner (RG Sureshkumar)

a few of them but I dont remember the specific ones

Agent

That's an honest answer, and it's also a gap worth naming. Since you can't point to which ones you checked, your record can't show a reader which figures you verified. Your research file also shows

sources S59–S80 accepted in a single message ("I accept all 22 proposed decision"). What were you relying on when you approved those 22?

12. Learner (RG Sureshkumar)

I have previous experience with Claude and trust many of the sources that Claude provides me with

Agent

Trust built on past experience is understandable, but it's trust in Claude, not in the sources. Your own project shows why that matters. The old calculator cited an EPA document for the TV figure, and that document doesn't contain it. S51 gives two phone-water values that don't match each other. A **citation** only tells you where to look; it doesn't show that the source supports the claim. The **responsibility** for what's published stays with you.

A practical habit for next time is to spot-check. Open the source behind each headline figure and find the number yourself.

Let's move to the spec. Take feature 2, the agent sessions. In plain language, how does the calculator turn a session into watt-hours?

13. Learner (RG Sureshkumar)

It relies on users' inputs and uses the information given by the user to calculate the energy usage

Agent

That's true of every feature, so let's get more specific. The session form asks for four kinds of tokens. What are they, and does each kind cost the same amount of energy?

14. Learner (RG Sureshkumar)

im not sure

Agent

Here's the short version. A token is a small chunk of text, roughly three-quarters of a word. Your spec counts four kinds:

- **Output** is what the model writes. It uses the chosen model's EcoLogits rate, the most expensive per token.
- **Fresh input and cache writes** are new text the model reads for the first time: about 2.1 Wh per 10,000 tokens (range 1–4).
- **Cache reads** are text the model has already read and reuses: 0.39 Wh per 10,000 (range 0–4).

The energy is each count multiplied by its rate, then added together. Your spec labels the cache-read range as "our estimate; no measurement exists." Why do you think that label matters to Jordan?

15. Learner (RG Sureshkumar)

he wants to ensure that he is only using the necessary amount of tokens for any job and not over using tokens

Agent

That could be one of Jordan's goals, but the label is doing something different. In your own acceptance check, cache reads make up 19.5 of the session's 46.7 Wh, and that is the part with no measurement behind it. Without the label, Jordan would read 46.7 Wh as measured fact. With it, they can see how much of the total rests on the project's own guess.

That's **transparency** about **uncertainty**. Does that change how you'd describe what the label is for?

16. Learner (RG Sureshkumar)

yes

Agent

Then put it in your own words: in one sentence, what should an employee take from a label saying a figure is "our estimate"?

17. Learner (RG Sureshkumar)

its about the user knowing his own usage

Agent

Not quite. It's less about knowing your usage and more about knowing **how far to trust the number**. "Our estimate" tells the employee that the figure is an **estimate**, not a measurement, so it could reasonably be much lower or higher. That's why it comes with a range.

Now for the spec itself. Feature 2's first acceptance check expects about 46.7 Wh for a particular session. Did you enter that session into the calculator yourself and see what came out?

18. Learner (RG Sureshkumar)

claude ran the acceptance checks

Agent

Understood. Your plan has a verification item, "User has checked feature behavior and calculations... independently of the agent," and it was ticked at your direction. Was there any check you did yourself, for any feature? Give me the input you entered and what you saw.

19. Learner (RG Sureshkumar)

I had claude open the application into the window that the feature uses and tested it

Agent

So the calculator was open in the preview pane. When you tested it, what did you do yourself? For example, did you log an image, change a region, or enter hours for a video call? And what did you expect to see?

20. Learner (RG Sureshkumar)

yes I logged a test usage and I expected to see the application show me how my energy usage compared to other things

Agent

That counts as a real check, and it's worth having done. It's one of three kinds, though:

1. **The control responds.** You logged something and the page changed. You did this one.
2. **The arithmetic is right.** One image should add 1.35 Wh, and you compare what shows up against that.
3. **The estimate is justified.** 1.35 Wh actually comes from the source.

An **acceptance check** is the second kind. You write the expected result down first, then compare. Which feature would be easiest to check at that second level?

21. Learner (RG Sureshkumar)

image logging

Agent

Agreed, it's the simplest. So set up the check: with the United States region selected, you log 3 images. Using your spec's per-image figure, what energy total should the tracker add?

22. Learner (RG Sureshkumar)

4.05 Wh

Agent

Correct: $3 \times 1.35 = 4.05$ Wh, with a range of about 0.26–34.5 Wh. That's a complete acceptance check: an input, an expected result worked out from the spec, and something to compare it against. Running it yourself in the tracker takes about a minute.

Now to revisions. During the build you decided that logged prompts should follow "Use my log to drive this page," just as images do, and that change was recorded in `spec.md`. Why the spec rather than the plan?

23. Learner (RG Sureshkumar)

because it changes what the calculator does

Agent

Exactly. That's the **specification versus implementation plan** distinction. The spec records what the calculator must do. The plan records the route: build order, dependencies, and checkpoints. Building feature 3 before feature 4 because the device breakdown sits inside the digital-day total is a plan decision.

On directing Claude: was there a moment when it misunderstood, drifted, or proposed something you rejected? What did you do?

24. Learner (RG Sureshkumar)

there were times when I rejected some of the suggestions that claude gave me

Agent

Can you give me one? Tell me what Claude suggested and why you turned it down.

25. Learner (RG Sureshkumar)

i dont remeber the specifices

Agent

[Tool: Bash — inspect the transcript format and search the research and specification transcripts for rejections] [Tool: Bash — list the learner's turns in transcripts/spec-2026-09-26_192752.md]

That's fine. I looked at your specification transcript instead. Most of your turns there are choices from Claude's lettered options, like "go with option a," often a dozen in a row. One decision was clearly your own at the start: "I want two totals side by side." When Claude laid out options a, b, and c, how did you decide between them?

26. Learner (RG Sureshkumar)

I picked the option that made the most sense

Agent

Let's test that on the decision that was yours. Why two totals side by side, rather than showing AI as a percentage of your whole digital day?

27. Learner (RG Sureshkumar)

to see how your usage compares based on different needs

Agent

That's part of it. There's also a stronger reason, and it's already in your spec. The AI total counts data centres only. The digital-day total also counts your device and home router. Because they're measured on different boundaries, a percentage like "AI is 4% of your digital day" would look precise but compare unlike things. It would also nudge the reader toward "AI is trivial."

That's **framing**: how you present a number shapes what people conclude from it. Can you think of another place in the calculator where the display could push someone toward a conclusion?

28. Learner (RG Sureshkumar)

the comparison charts with everyday items such as burgers

Agent

Right, that's the clearest case. How could the burger comparison push someone toward "AI is harmless" or "AI is harmful"? Think about the old figure compared with the one you have now.

29. Learner (RG Sureshkumar)

what about it

Agent

Here's the concrete version. The old methodology quoted a burger at about 1,700 L of water. Most of that is rain falling on fields, while the AI figures count only water actually used up for cooling and power generation. Next to 1,700 L, a few millilitres of AI water looks negligible, which pushes toward "harmless." Your corrected blue-water figure is about 62 L, with a range of roughly 8–226 L, so both sides now measure the same kind of water.

That's what makes a **comparison** meaningful: compatible quantities and stated assumptions. Did you check the burger figure in the running calculator after the change?

30. Learner (RG Sureshkumar)

no, claude checked it

Agent

That's a pattern worth naming plainly. Your plan's verification section says you checked behaviour "independently of the agent" and confirmed the citations, and those boxes were ticked at your direction. From what you've described, Claude ran the acceptance checks and you tested that the log responds. **Verification** means comparing the work against the requirements yourself, not relying on a report that it's done.

The fix is small. Pick two or three acceptance checks, like your 3-image one, and run them yourself before ticking anything.

Moving on: the brief requires the features to serve Alex, Jordan, and Robin. Which feature does the most for Robin, and why?

31. Learner (RG Sureshkumar)

like for like comparisons, because Robin is skeptical

Agent

That holds up. Remember, though, that "Robin is skeptical" comes from a fictional profile you were given, not from talking to real employees. **User research** means observing or asking actual people. Your research looked at the activities the profiles describe, which is useful, but it doesn't show what real employees at the company need.

Robin also spends a lot of time in video meetings. Which feature covers that, and what does it add to the calculation?

32. Learner (RG Sureshkumar)

The digital day builder, it adds video call hours by device

Agent

Right. Each hour counts the device's power, the home router, and the network's data. Most of those figures come from S25, the Carbon Trust's 2021 study, which your spec notes was seed-funded by Netflix. How should that funding affect how much you trust S25?

33. Learner (RG Sureshkumar)

it could be biased since netflix funded it

Agent

Yes, funding is a real reason for caution, but it doesn't settle the question on its own. **Source credibility** also depends on the method, whether it's disclosed, and whether independent sources agree. Here, the IEA's Kamiya (S21) separately found that the viewing device dominates streaming energy, which supports S25's main finding. Funding is a reason to look for that kind of corroboration, not a reason to discard the study automatically.

Now feature 4, device manufacturing. In plain terms, how does a phone's manufacturing carbon get into a single day's total?

34. Learner (RG Sureshkumar)

it is accrued thoghout its usage

Agent

That's the idea: it's spread across the time you own it. So if your phone's manufacturing carbon is about 42 kg and you keep it 3 years, you get about 38 g per day. What happens to that daily figure if you keep the phone 6 years instead?

35. Learner (RG Sureshkumar)

it gets cut in half

Agent

Correct: about 19.2 g a day. That's one of your spec's acceptance checks, and you just worked it out yourself.

Now, the 42 kg comes from one iPhone model's report from Apple, and the daily share depends on a guessed number of years. Your acceptance checks still quote figures like "38.4 g" and "46.65 Wh." Does showing that decimal place tell employees anything true?

36. Learner (RG Sureshkumar)

it just shows the employees more detail

Agent

It looks like more detail, but the extra digit isn't backed by the evidence. When the inputs are one phone model and a guess about how long someone keeps it, "38.4 g" implies you know the answer to within a tenth of a gram. You don't. That's **false precision**. "About 40 g a day" would be more honest, and on the page, a range does that job better.

Did you notice how the page itself rounds figures, or is that unchecked?

37. Learner (RG Sureshkumar)

no i didnt check that

Agent

[Tool: Bash — search assets/js/.js for number formatting; fmtNum in core.js uses 1 decimal at ≥ 10 , 2 decimals at ≥ 1 , 3 decimals at ≥ 0.01]*

I checked the code but didn't run the page. In assets/js/core.js:41, the shared number formatter keeps about three significant figures at every scale: "38.4 g", "1.35 Wh", "0.513 g". So an image's carbon would show as 0.513 g next to a range of 0.033–4.37 g. The range tells the honest story, while the central value suggests more certainty than the evidence supports.

That's a concrete fix to raise for a later round, not now. Next, the **methodology** section: did you read what the page tells employees about how the figures were produced?

38. Learner (RG Sureshkumar)

no i did not read it closely

Agent

That matters, because the **methodology** section is where a skeptical reader like Robin goes to judge whether to trust the numbers. It says what was measured, how figures were combined, and what was left out. Claude rewrote it for feature 5, so right now it's text that no human on the project has reviewed. It's worth reading once, top to bottom, as Robin would.

Here's a hypothetical. Suppose a new study measures cache reads directly and finds they use almost no energy, close to 0. What would you reconsider in feature 2, and why?

39. Learner (RG Sureshkumar)

no idea

Agent

Let's break it down. In your acceptance check, cache reads were 19.5 Wh of the session's 46.7 Wh. If the new study showed they're nearly zero, what would happen to that session's total?

40. Learner (RG Sureshkumar)

it would be unchanged

Agent

Not quite. The cache-read part would drop from 19.5 Wh to almost nothing, so the session total would fall from about 46.7 Wh to about 27 Wh, roughly 40% lower. The better **evidence** would replace your own guess, and the "our estimate" label could come off.

So you'd reconsider the cache-read rate and range in the spec, then re-run acceptance check 1. Does that sequence make sense?

41. Learner (RG Sureshkumar)

yes it does

Agent

[Tool: WebFetch — <https://arxiv.org/html/2311.16863v3> : image generation mean 2.907 kWh per 1,000 inferences (SD 3.31); least efficient model stable-diffusion-xl-base-1.0 at 1,594 g CO₂eq per 1,000, "equivalent to approximately 11.49 kWh"; 8 NVIDIA A100-SXM4-80GB GPUs on AWS us-west-2] [Tool: WebFetch — same page, asked about the median: the tool reported "image generation (1.35 kWh)" as the median across task categories]

Before we wrap up, I went back to the concept statement, which is about verifying the AI's account against the source itself. Since you relied on Claude's summary of S1, I checked it: I fetched the arXiv paper, read through a summarising tool rather than line by line. It does give 1.35 kWh per 1,000 image generations as the median, measured on NVIDIA A100 GPUs on AWS. So 1.35 Wh per image holds up.

The 11.49 figure is less clear. The fetch tied it to a carbon figure for the least efficient model, and I couldn't tell whether 11.49 kWh is printed in the paper or worked out from that carbon number. That's unverified until someone reads Table 2 directly.

One more question: did you open the calculator yourself, outside the preview pane, for example by double-clicking `index.html`?

42. Learner (RG Sureshkumar)

yes i did

Agent

Good. That's your own check that it runs locally.

Here's my recap.

What we covered - All five features: images and agent sessions for professional AI use, the digital day and device manufacturing for broader digital life, and like-for-like comparisons as the fifth. We also covered how they map onto Alex, Jordan, and Robin. - The route through research, spec, plan, build, and verification, plus the Git checkpoints.

Where your reasoning was solid - "Two totals side by side" was your own decision, and it's the right one. The totals are measured on different boundaries, so showing AI as a percentage would have

misled people. - You placed the "Use my log" change in the spec because it changes what the calculator does. That's the spec-versus-plan distinction, applied correctly. - You worked out two acceptance checks yourself: 3 images → 4.05 Wh, and a phone kept 6 years → half the daily share. - You opened the calculator locally yourself, and you tested that logging responds.

What became clearer - **Audience** and **user need** mean the employees' needs, not your preference. The profiles are fictional, so they aren't user research. - A citation shows where to look; a summary isn't the source. Funding calls for looking for corroboration, not for automatically rejecting a study. - "Our estimate" labels are about how far to trust a number, and extra decimal places can create false precision.

Worth tightening - **Verification**: the "checked independently" and "confirmed citations" boxes were ticked, but Claude ran the acceptance checks, and you relied on Claude's summaries for most sources, including the 22 accepted in one message. Next time, run two or three acceptance checks yourself and open the source behind each headline figure before ticking anything. - **Directing the agent**: most spec decisions were "go with option a." Before choosing, ask Claude what each option would mean for an employee, or propose your own option. - **Unresolved**: the methodology text hasn't been read by a person, the S1 11.49 figure is unconfirmed, and the number formatter in `assets/js/core.js:41` displays three significant figures.

The bigger picture: Project 2 took a client need all the way through research, feature choices, a spec, a plan, a build, and verification. You finished that full loop, and the one decision you drove yourself shaped the whole design. On the next project, the habit to carry forward is to own the checking the way you owned that decision.

43. Learner (RG Sureshkumar)

save the transcript of this debrief

Agent

[Tool: Bash — read transcripts/README.md and the Unit 1 debrief header; system clock: 2026-09-26T23:48:05-05:00] [Tool: Write — transcripts/debrief-2026-09-26_234805.md]